INCH-POUND

MIL-DTL-17999D (MR)
14 July 2025
SUPERSEDING
MIL-M-17999C (MR)
1 December 1972

DETAIL SPECIFICATION

METAL, EXPANDED, ALUMINUM

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers aluminum expanded metal.

1.2 Classification. Expanded metal should be manufactured of the following class and alloys, as specified (See 6.2):

1.2.1 Class 1. As expanded.

1.2.2 Class 2. Expanded and flattened.

1.2.3 Alloys. The following Aluminum Association designated alloys should be used.

1.2.3.1 Alloy 3003.

1.2.3.2 Alloy 5005.

1.2.3.3 Alloy 5052.

1.2.3.4 Alloy 6061.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-L-10547	-	Liners, Case, Waterproof

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

(Copies of this documents are available online at https://quicksearch.dla.mil/).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE ALUMINUM ASSOCIATION, INC.

Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org).

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes
ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products

(Copies of these documents are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM B209/B209M	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate Spectrometry

(Copies of these documents are available online at https://www.astm.org/).

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NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS

EMMA 557 - Standards For Expanded Metal

(Copies of these documents are available online at https://www.naamm.org).

3. REQUIREMENTS

3.1 Materials. Unless otherwise specified in the contract or purchase order (see 6.2), expanded metal and flattened expanded metal shall be produced from aluminum alloy 3003-H14, 5005-H34, 5052 in O or H32 tempers, or 6061-T6 sheet conforming to the chemical compositions defined by the Aluminum Association Teal Sheets, "International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys" listed in table I and within those limits. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

TABLE I. Chemical composition, weight percent. 1/ 2/

ELEMENTS	SYMBOL	3003 ALLOY	5005 ALLOY	5052 ALLOY	6061 ALLOY
Silicon	Si	0.6	0.30	0.25	0.40 - 0.8
Iron	Fe	0.7	0.7	0.40	0.7
Copper	Cu	0.05 - 0.20	0.20	0.10	0.15 - 0.40
Manganese	Mn	1.0 - 1.5	0.20	0.10	0.15
Magnesium	Mg	---	0.50 - 1.1	2.2 - 2.8	0.8 - 1.2
Chromium	Cr	---	0.10	0.15 - 0.35	0.04 - 0.35
Zinc	Zn	0.10	0.25	0.10	0.25
Titanium	Ti	---	---	---	0.15
Other, max. Each	---	0.05	0.05	0.05	0.05
Other, max. Total 3/	---	0.15	0.15	0.15	0.15
Aluminum	Al	Remainder	Remainder	Remainder	Remainder

1/ Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those "others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.2 Manufacture. Expanded metal shall be sourced from sheet aluminum manufactured in accordance with ANSI H35.2 and ASTM B209 in thicknesses corresponding to table II, as specified in (see 6.2). Unless otherwise specified in the contract or purchase order, the aluminum shall be sheared so that each sheet will be expanded into uniform diamond-shaped openings, the longer diagonals of which shall be parallel to the length direction of the sheet. Each opening shall be integral with adjoining openings by means of unsheared bonds (bridges) of the original sheet.

3.2.1 Class 1. The strands which form the sides of the openings of class 1 expanded metal shall be straight, and shall be rectangular in cross section.

3.2.2 Class 2. Class 2 flattened expanded metal shall have the strands and bonds in the same plane as a result of passing through flattening rolls.

3.3 Weights, dimensions and tolerances.

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3.3.1 Sheet size. Unless otherwise specified in the contract or purchase order, class 1 expanded metal and class 2 flattened expanded metal shall be furnished in sheets 4 feet wide by 8 feet long.

3.3.2 Class 1. Class 1 expanded metal shall conform to the weights, dimensions, and tolerances specified in table II unless otherwise specified in the contract or purchase order. Additional expanded metal configurations may be found within EMMA 557, "Standards for Expanded Metal".

3.3.3 Class 2. Before flattening, class 2 expanded metal shall conform to table II. After flattening, thickness of sheet may be reduced not more than 25 percent of the original thickness as a result of passing through flattening rolls. Weight per square foot may be reduced not more than 5 percent as a result of elongation produced by passing through flattening rolls.

TABLE II. Mesh sizes, thickness of sheet, and weight of class 2 expanded metal before flattening.

Commercial style number	Size of mesh, inches			Nominal thickness of manufactured sheet, inch	Width of strand, inch	Weight per square foot, pounds 2/
	Nominal width of opening	Dimensions, center to center of bridges 1/				
		width	length			
1/2 - No. .050	0.5	0.48	1.2	0.051	0.079 - 0.095	0.27
1/2 - No. .080	0.5	0.48	2.0	0.081	0.079 - 0.097	0.44
3/4 - No. .050	0.75	0.90	2.0	0.051	0.092 - 0.114	0.17
3/4 - No. .080Lt	0.75	0.90	2.0	0.081	0.093 - 0.130	0.32
3/4 - No. .080Hy	0.75	0.90	2.0	0.081	0.138 - 0.172	0.41
3/4 - No. .125	0.75	0.91	3.0	0.125	0.138 - 0.172	0.65
1-1/2 - No. .080	1.5	1.34	3.0	0.081	0.107 - 0.133	0.22
1-1/2 - No. .125	1.5	1.34	1.2	0.125	0.134 - 0.168	0.43

1/ A tolerance of plus or minus 10 percent is permitted in dimensions, center to center.
2/ A variation in weight per square foot of 5 percent is permissible, based on the weight of any sheet or bundle.

3.4 Workmanship. When visually inspected, the expanded metal strands shall be of substantially uniform width and thickness and shall be smooth and free from sharp edges. Metal shall be free from waves, buckles, burns, smut, or corrosion. Broken strands, weld-repaired strands, irregular shaped openings, and any other defects, which may affect serviceability shall not be permitted.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirement, as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Lot. A lot shall consist of all expanded metal sheets of the same class, alloy, and size, and manufactured under the same conditions, submitted for inspection at one time.

4.3 Sampling for examination. Samples for examination shall be selected in accordance with MIL-STD-1916, or ANSI/ASQ Z1.4 inspection level II, 1.5 percent AQL.

4.4 Examination.

4.4.1 Visual and dimensional. Each of the sample expanded metal sheets selected in accordance with 4.3 shall be visually and dimensionally examined to verify compliance with this specification.

4.4.2 Preparation for Shipment. Examination of the packaging, packing, and marking for shipment shall be made for conformance to the requirements of section 5.

4.5 Rejection. Any expanded metal sheet in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective expanded metal sheets in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.1.1 Packing details. For additional considerations for packing expanded aluminum see 6.5.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Expanded metal covered by this specification is intended for machinery guards, ventilating screens, and other general uses.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Class, alloy and temper required (see 1.2 and 3.1).
c. When other than uniform diamond-shaped openings are required.

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d. Commercial style number.
e. when special size sheets are required.
f. Level of packaging and packing required (see 5.1)
g. Special marking, when required (see 5.2)

6.3 Alloy density. The density of the specified alloys listed in this specification are as follows:

AA3003 = 0.099 lb/in^3 (2.73 g/cm3).
AA5005 = 0.098 lb/in^3 (2.70 g/cm3).
AA5052 = 0.097 lb/in^3 (2.68 g/cm3).
AA6061 = 0.098 lb/in^3 (2.70 g/cm3).

6.4 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.5 Packing notes. For optimized shipment and potential extended storage, expanded aluminum, segregated for size and class, should be stacked and wrapped in waterproof barrier materials and packed for shipment in secured lifts. Secured lifts for expanded metal should be constructed using lumber grades sufficient to bear the load with constructed skids running lengthwise or crosswise with respect to the load. The loads should be secured in tension with a minimum of five straps: three girthwise and two lengthwise. The strapping in the direction of the skids should be placed over the skids and securely stapled to the skids with corrosion resistant staples. Strapping should be 1.25 by 0.035-inch flat band or 8 gage high tensile wire. The gross weight of secured lifts should not exceed 3000 pounds. In addition to any special marking required in the contract or purchase order, marking of the expanded metal lifts should be in accordance with MIL-STD-129.

6.6 Subject term (key word) listing.

Cages	Grating	Lath	Mesh
EMI/RFI	Grids	Lattice	Shielding
Faraday Cage	Grills	Matting	Strainers

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
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Preparing activity:
ARMY - MR

(Project 9535-2025-009)

Review activities:
Army - AT, ME
DLA - CQ

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